

$Q = 200\ l/s = 0.2\ m^3/s$

1. $\Sigma F_x = 0 \Rightarrow -R_x + P_A A_A + \rho Q V_A = 0$

$\Sigma F_y = 0 \Rightarrow P_B A_B + \rho Q V_B - R_y = 0$

$V_A = \frac{0.2}{\frac{\pi \times 0.4^2}{4}} = 1.59\ m/s \qquad V_B = \frac{0.2}{\frac{\pi \times 0.2^2}{4}} = 6.37\ m/s$

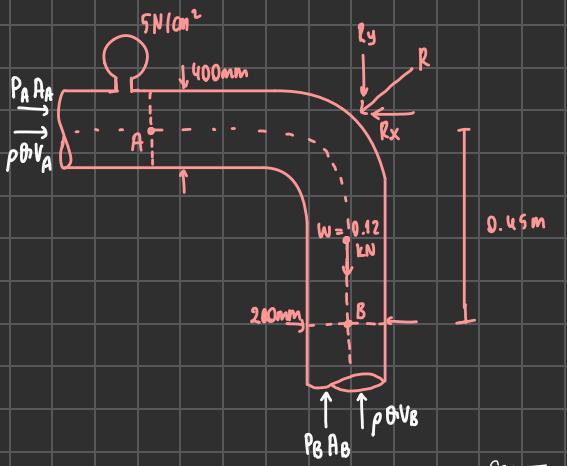
2. $\frac{P_A}{\gamma} + \frac{V_A^2}{2g} + z_A = \frac{P_B}{\gamma} + \frac{V_B^2}{2g} + z_B$

$\frac{50}{9.81} + \frac{1.59^2}{2\times 9.81} + 0.45 = \frac{P_B}{9.81} + \frac{6.37^2}{2\times 9.81} \Rightarrow P_B = 35.39\ kN/m^2$

$\gamma = 9.81\ kN/m^3 = 9810\ N/m^3$

3. $R_x = 6.6\ kN \leftarrow$

$R_y = 2.26\ kN \downarrow$



$\theta_A = \theta_B = A_A V_A = A_B V_B \rightarrow V_A \times (0.04)^2 = V_B \times (0.02)^2$

$4V_A = V_B$

$\Sigma F_x = 0$

$\rho Q V_A + P_A A_A = R_x$

$\Sigma F_y = 0$

$\rho Q V_B + P_B A_B - W = R_y$

$\frac{P_A}{\gamma} + \frac{V_A^2}{2g} + z_A = \frac{P_B}{\gamma} + \frac{V_B^2}{2g} + z_B$

$\frac{50000}{9.81 \times 1000} + \frac{(1.59)^2}{2\times 9.81} + 0.45 = \frac{P_B}{9.81\times 1000} + \frac{(6.37)^2}{2\times 9.81}$

$P_B = 35.39\ kN/m^2$